

GIT & GITHUB

What is Git?

Git is a distributed **version control system used to track changes in source code**. It helps developers manage code, collaborate with teams, and maintain project history.

Features of Git

- Tracks code changes
- Supports collaboration
- Branching and merging
- Maintains version history

What is GitHub?

GitHub is a cloud-based platform used to host Git repositories online. It helps developers store, manage, and collaborate on projects.

Features of GitHub

- Remote repository hosting
- Team collaboration and Pull requests
- Issue tracking
- CI/CD integration

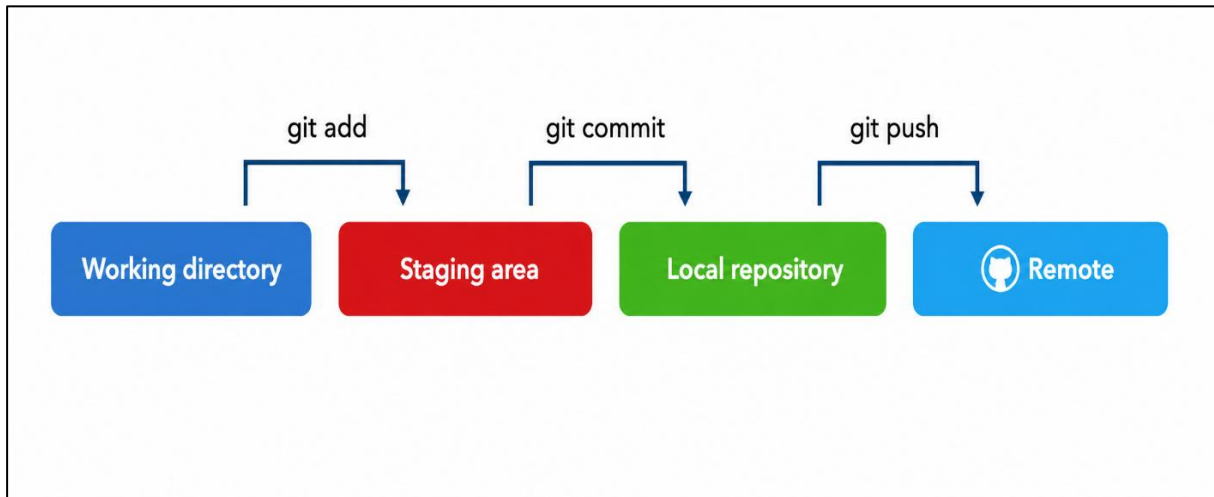
Difference Between Git and GitHub

Git	GitHub
Version control tool	Cloud hosting platform
Works locally	Works online
Tracks code changes	Stores repositories
Command-line based	Web-based interface

Basic Git Workflow

1. Create or modify files
2. Add files to staging area
3. Commit changes to local repository
4. Push code to remote repository

GIT ARCHITECTURE DIAGRAM



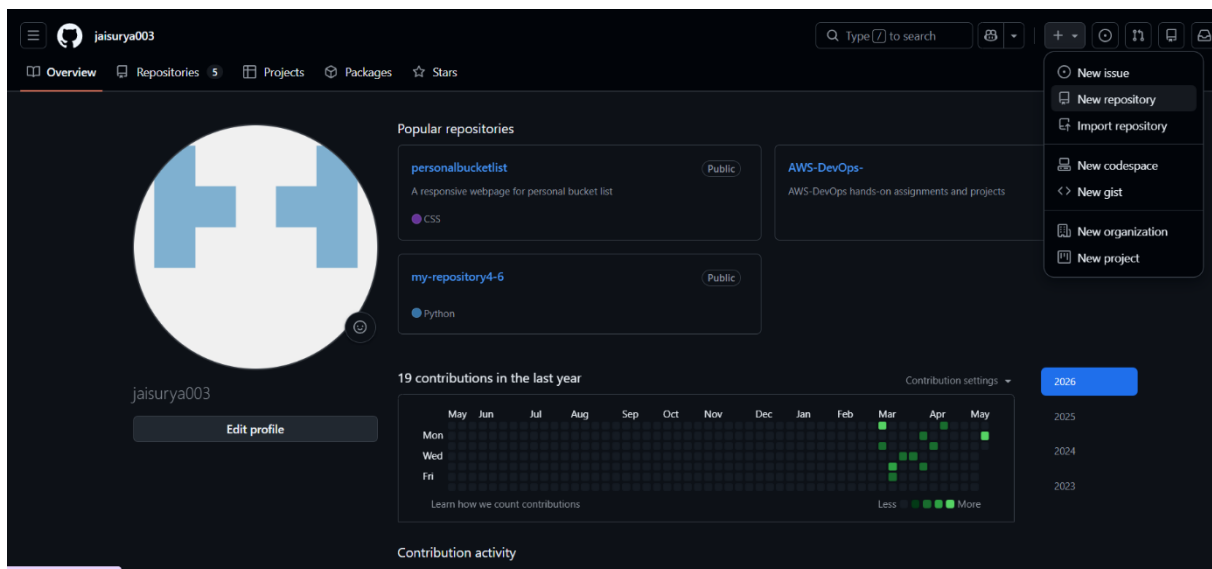
Git Commands

Command	Definition
git clone	Used to copy a remote GitHub repository to local system
cd	Used to move into a directory or folder
ls -la	Used to view all files and folders including hidden files
touch	Used to create a new file
vim	Used to open and edit a file in Vim editor
cat	Used to display the contents of a file
git status	Used to check the current status of repository files
git status -s	Used to view repository status in short format
git add	Used to add files to staging area
git commit -m	Used to save changes to local repository with a message
git log	Used to view detailed commit history
git log --oneline	Used to view commit history in one line format
git push origin main	Used to push local commits to GitHub repository

HANDS-ON FOR GIT AND GITHUB

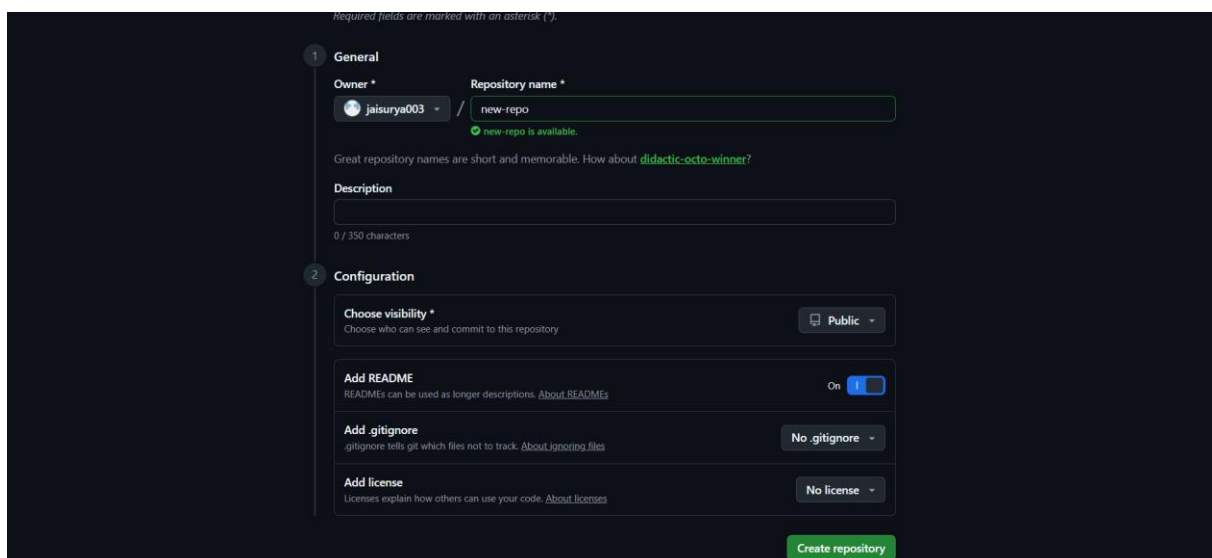
Screenshot: Create New Repository in GitHub

- Open GitHub
- Click New
- Create new repository



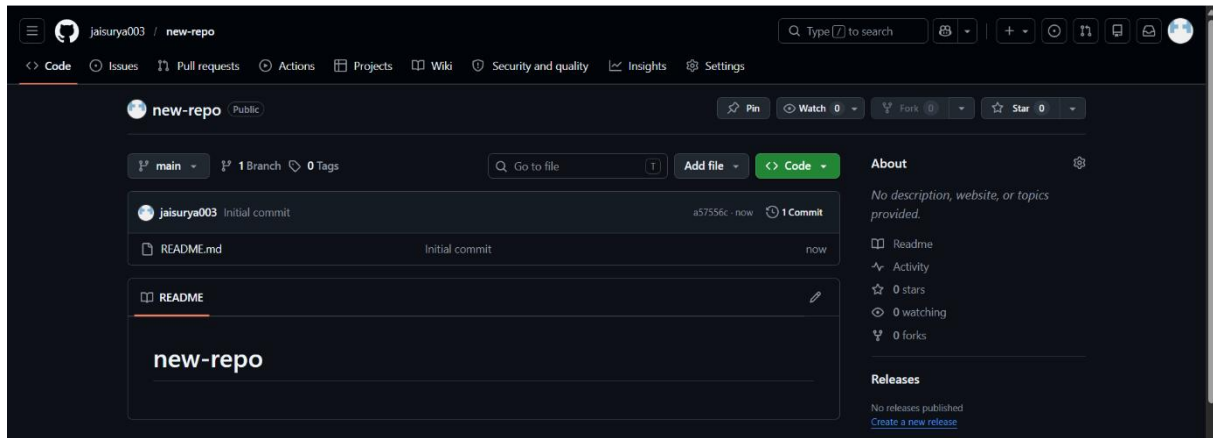
Screenshot: Enter Repository Details

- Enter repository name
- Enable README file
- Click Create repository



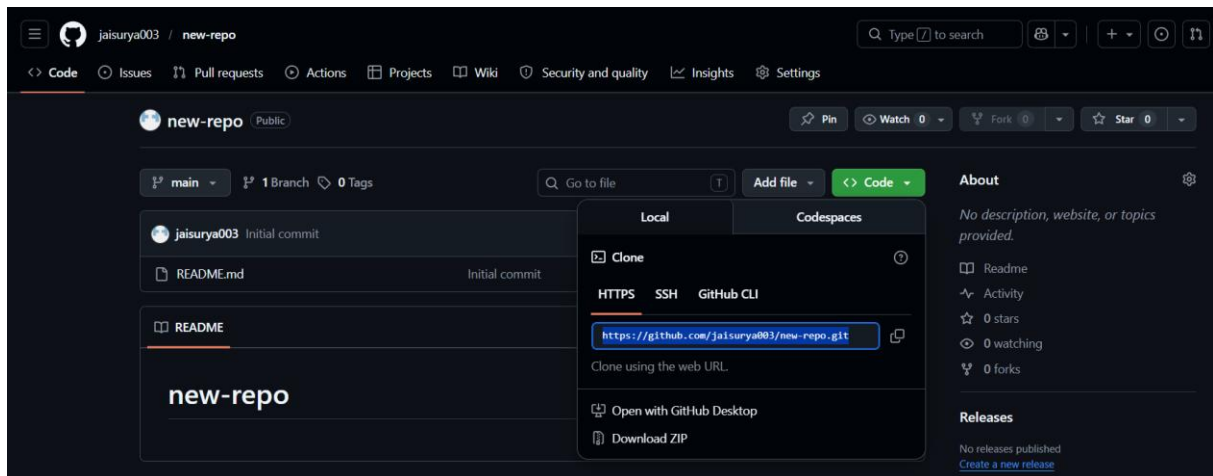
Screenshot: Repository Created Successfully

- New repository created
- README added as initial commit



Screenshot: Copy Repository HTTPS Link

- Go to Code
- Copy HTTPS repository link



Screenshot: Clone GitHub Repository

- Open Git Bash
- Execute `cd Downloads` (to move into Downloads folder)
- Execute `git clone "copied link"` (to clone the GitHub repository)
- Execute `cd new repo` (to move into the repository folder)
- Execute `ls -la` (to view all files and folders)

```
MINGW64:/c/Users/JAISURYA/downloads/new-repo

JAISURYA@LAPTOP-G96D93Q4 MINGW64 ~
$ cd downloads

JAISURYA@LAPTOP-G96D93Q4 MINGW64 ~/downloads
$ git clone https://github.com/jaisurya003/new-repo.git
Cloning into 'new-repo'...
remote: Enumerating objects: 3, done.
remote: Counting objects: 100% (3/3), done.
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (3/3), done.

JAISURYA@LAPTOP-G96D93Q4 MINGW64 ~/downloads
$ cd new-repo

JAISURYA@LAPTOP-G96D93Q4 MINGW64 ~/downloads/new-repo (main)
$ ls -la
total 165
drwxr-xr-x 1 JAISURYA 197121  0 May 12 09:23 ./
drwxr-xr-x 1 JAISURYA 197121  0 May 12 09:23 ../
drwxr-xr-x 1 JAISURYA 197121  0 May 12 09:23 .git/
-rw-r--r-- 1 JAISURYA 197121 10 May 12 09:23 README.md

JAISURYA@LAPTOP-G96D93Q4 MINGW64 ~/downloads/new-repo (main)
$
```

Screenshot: Create Python File

- Execute touch one.py (to create a new Python file)
- Execute vim one.py (to open the file in Vim editor)

```
JAISURYA@LAPTOP-G96D93Q4 MINGW64 ~/downloads/new-repo (main)
$ touch one.py

JAISURYA@LAPTOP-G96D93Q4 MINGW64 ~/downloads/new-repo (main)
$ vim one.py
```

Screenshot: Save File in Vim Editor

- Press Esc :wq (to exit insert mode and o save and quit the file)

```
MINGW64:/c/Users/JAISURYA/downloads/new-repo

my first file
~
~
~
~
~
```

Screenshot: Check Git File Status

- Execute cat one.py (to display file content)
- Execute git status (to check repository status)
- Execute git status -s (to view short status output)
- Red color indicates untracked file status

```
JAISURYA@LAPTOP-G96D93Q4 MINGW64 ~/downloads/new-repo (main)
$ cat one.py
my first file

JAISURYA@LAPTOP-G96D93Q4 MINGW64 ~/downloads/new-repo (main)
$ git status
On branch main
Your branch is up to date with 'origin/main'.

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    one.py

nothing added to commit but untracked files present (use "git add" to track)

JAISURYA@LAPTOP-G96D93Q4 MINGW64 ~/downloads/new-repo (main)
$ git status -s
?? one.py
```

Screenshot: Add File to Staging Area

- Execute git add one.py (to add file to staging area)
- Execute git status -s (to view short status output)
- Green color indicates staged file status

```
JAISURYA@LAPTOP-G96D93Q4 MINGW64 ~/downloads/new-repo (main)
$ git add one.py
warning: in the working copy of 'one.py', LF will be replaced by CRLF the next time Git touches it

JAISURYA@LAPTOP-G96D93Q4 MINGW64 ~/downloads/new-repo (main)
$ git status -s
A  one.py

JAISURYA@LAPTOP-G96D93Q4 MINGW64 ~/downloads/new-repo (main)
$ git status
On branch main
Your branch is up to date with 'origin/main'.

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
    new file:   one.py

JAISURYA@LAPTOP-G96D93Q4 MINGW64 ~/downloads/new-repo (main)
```

Screenshot: Commit Changes to Repository

- Execute git commit -m "first commit" (to save changes in local repository)
- Execute git log (to view commit history)
- Execute git log --oneline (to view commit history in one line format)

```

JAISURYA@LAPTOP-G96D93Q4 MINGW64 ~/downloads/new-repo (main)
$ git commit -m "first commit"
[main 063b39f] first commit
1 file changed, 1 insertion(+)
create mode 100644 one.py

JAISURYA@LAPTOP-G96D93Q4 MINGW64 ~/downloads/new-repo (main)
$ git log
commit 063b39ff3986ee6be518aed283ca0475dd4dd316 (HEAD -> main)
Author: jaisurya003 <jaisuryajai003@gmail.com>
Date: Tue May 12 09:31:21 2026 +0530

    first commit

commit a57556c297cd71f825a0feeb9cc555aa94c83fe7 (origin/main, origin/HEAD)
Author: jaisurya003 <146946666+jaisurya003@users.noreply.github.com>
Date: Tue May 12 09:10:52 2026 +0530

    Initial commit

JAISURYA@LAPTOP-G96D93Q4 MINGW64 ~/downloads/new-repo (main)
$ git log --oneline
063b39f (HEAD -> main) first commit
a57556c (origin/main, origin/HEAD) Initial commit

JAISURYA@LAPTOP-G96D93Q4 MINGW64 ~/downloads/new-repo (main)
$

```

Screenshot: Push Changes to GitHub

- Execute git push origin main (to push local commits to GitHub repository)

```

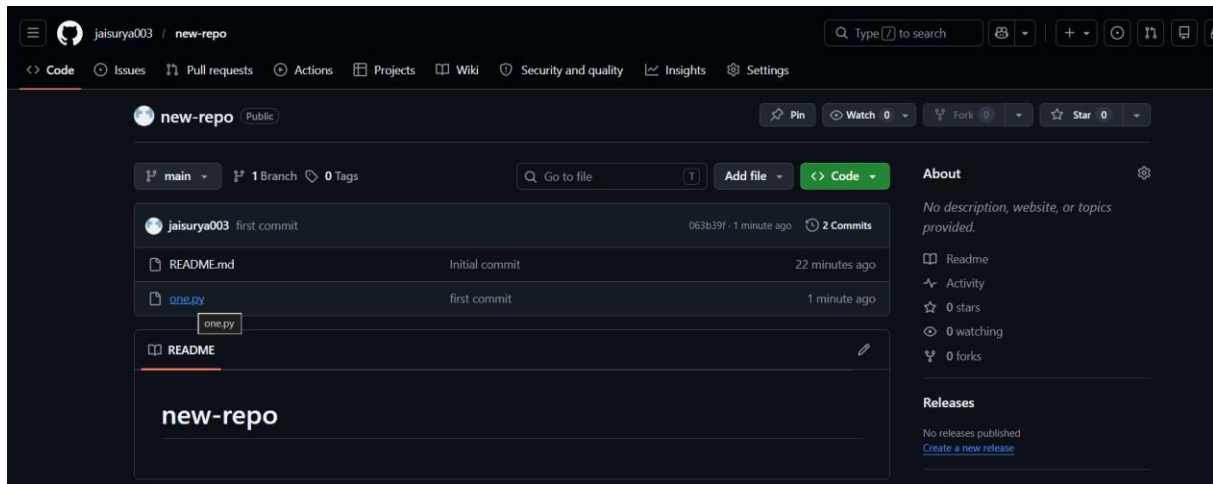
JAISURYA@LAPTOP-G96D93Q4 MINGW64 ~/downloads/new-repo (main)
$ git push origin main
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 12 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 284 bytes | 284.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/jaisurya003/new-repo.git
    a57556c..063b39f  main -> main

JAISURYA@LAPTOP-G96D93Q4 MINGW64 ~/downloads/new-repo (main)
$

```

Screenshot: Verify Changes in GitHub

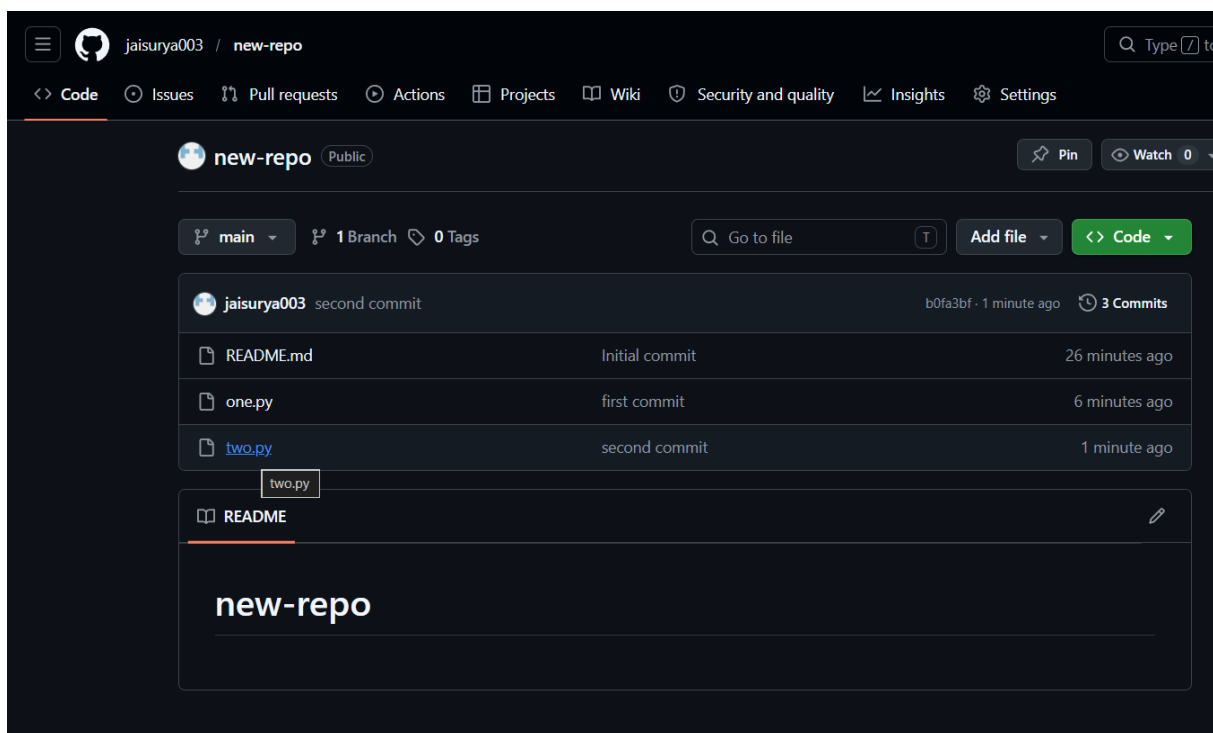
- Go to GitHub repository
- Refresh the page
- View one.py added in first commit



Screenshot: Repeat the same process for the second file

```
80 touch two.py
81 vim two.py
82 cat two.py
83 git status -s
84 git add two.py
85 git status -s
86 git commit -m "second commit"
87 git log --oneline
88 git push origin main
89 history
```

JAISURYA@LAPTOP-G96D93Q4 MINGW64 ~/downloads/new-repo (main)
\$ |



Screenshot: Multiple Files Added to Repository

- Ten files added to the GitHub repository
- Changes successfully pushed to remote repository

